

ResearchPilot AI – Problem Statement

Background

Researchers, students, and academicians spend significant time searching for relevant literature, reading lengthy research papers, comparing multiple studies, extracting important insights, and managing references. Existing search engines and AI chatbots often provide fragmented assistance, lack contextual understanding across documents, and may generate inaccurate or unsupported responses.

Problem Statement

The objective is to develop **ResearchPilot AI**, an intelligent Agentic AI-powered research assistant that automates the end-to-end research workflow. The system should search scholarly literature, summarize research papers, answer questions from uploaded documents, compare multiple papers, identify research gaps, generate research ideas, and assist with citation-aware research using Retrieval-Augmented Generation (RAG) and IBM Granite models.

Objectives

- Reduce the time required for literature review.
- Provide accurate, context-aware answers using RAG.
- Summarize and compare multiple research papers.
- Identify research gaps and suggest future research directions.
- Support document-based question answering.
- Deliver secure, scalable, and responsible AI using IBM watsonx.ai and IBM Granite.

Expected Outcome

ResearchPilot AI will function as an intelligent digital research assistant that improves research productivity by automating repetitive tasks, enhancing literature analysis, and helping users discover meaningful insights faster.